

LSMS Research Impact Tracking Pipeline

Full documentation and user guide

Executive Summary

The pipeline answers one question at scale: which published research actually uses LSMS data? It searches OpenAlex, the largest open index of scholarly work, for 109 survey keywords across 10 survey families, and then decides, paper by paper, whether the match reflects genuine use of the data or only a passing mention. **The central design choice is that those are two separate questions, scored separately.**

Earlier versions collapsed them into a single relevance score, which let a paper pass on name recognition alone. A literature review naming the survey three times, written by a World Bank author, outscored a real empirical study. The current version scores an **identity** axis (is this genuinely one of our surveys?) and a **use** axis (did the authors work with the microdata?) independently, and a paper reaches the results only if it clears both. This makes “mentions the survey but never uses the data” structurally unable to pass, which is what the previous design could not guarantee.

Two further decisions matter for accuracy. Searches run unstemmed, because OpenAlex by default treats LSMS and LSM as the same word and returns the entire land-surface model literature: 13,506 results for a single fiscal year against 698 unstemmed. And the phrase lists that detect data use were selected by measurement against a 36-paper review set rather than by intuition, which removed three phrases that turned out to indicate citation rather than use. **Against that review set the pipeline now agrees with manual assessment on 22 of 28 papers, at a cost of roughly \$1.60 per full run.**

The remaining disagreements are understood rather than mysterious. They divide into papers whose survey name appears only in the reference list, and papers whose data use is declared in a table or dataset section that OpenAlex does not index. Both limits are set out in section 2.10.

At a glance	Summary
What it measures	TOTAL, FLOW and SHARE: papers using LSMS data, new papers per WBG fiscal year, and the share with African-affiliated authors
Where it searches	OpenAlex, covering journal articles, working papers, preprints and grey literature, searched against title, abstract and full text
How it matches	109 keywords in 10 survey families, each on one of three tiers according to how ambiguous the name is on its own
How it decides	Two independent axes, identity and use, each scored on a 2-point threshold; a paper must clear both
What it produces	One Excel workbook: results, analysis with charts, keyword register, search audit trail, and every rejected paper with the reason
What it costs	About \$1.60 per full run; free tier allows roughly two runs per day

1. Project Description

This pipeline is a fully automated Python program. It finds every research paper that uses an LSMS-produced dataset, checks how strong the evidence is for each match, and exports a structured Excel workbook. The entire process, including discovery, validation, and export, takes about five minutes with no manual work required. The program runs through a custom-built interface. A user enters a free API key, clicks one button, and receives a finished excel workbook with the list of all Papers.

1.1 Original Goal

The original goal of this project was to collect more objective metrics on the impact of LSMS. Primarily, to replace manual tracking with a fast, repeatable process that reliably measures the three requested metrics: TOTAL, FLOW, and SHARE.

Metric	Definition
TOTAL	The full count of papers using an LSMS dataset, across all years.
FLOW	The number of new papers per World Bank fiscal year (FY = July 1 to June 30).
SHARE	The percentage of papers with an African-affiliated author.

1.2 Final Version

The final output goes beyond the three requested numbers. Every run produces a complete workbook that includes:

- A full paper-by-paper record (**Papers excel sheet**), including:
 - title, DOI, authors, institutional affiliations, and publication date (with FY categorization) for every paper,
 - several (30+) other collected metrics.

- A confidence breakdown showing how each paper was matched, and its identity and use scores.
- Automatically created analysis and visualization (**Analysis excel sheet**), including:
 - metrics and charts on Total, Flow and Share,
 - a chart about publication-type breakdown: peer-reviewed papers, working papers, and reports.
- A full audit trail of every search run and what it returned.
- A **custom interface** that makes the program easy to run for anyone.
- All this using an API configuration that makes it possible to run the program **free** up to twice a day.

2. Documentation

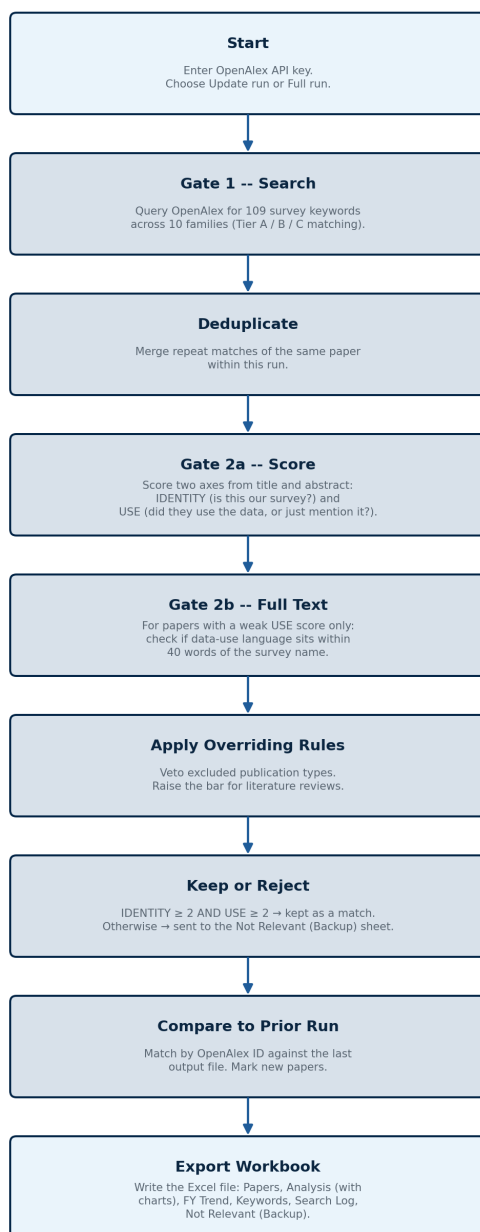
2.1 How the pipeline works

The pipeline runs in two steps. Everything else in this section belongs to one or the other, so it is worth separating them clearly before going further.

Step one (Gate 1) searches for the survey names and collects every paper that mentions one. It is deliberately generous. Missing a paper here is permanent, whereas collecting one that turns out to be irrelevant costs nothing but a later rejection.

Step two (Gate 2) examines each collected paper and decides whether it actually used the survey data. This is where most of the work happens, because **most papers that mention a survey never use it**. They cite it, credit it, or list it as related work.

The diagram below lays out that same sequence step by step, from entering the API key to the finished workbook:



2.2 Where the papers come from

The pipeline searches OpenAlex, a free, comprehensive scholarly index that covers journal articles, working papers, preprints, and grey literature (arXiv, SSRN, NBER, and similar repositories).

2.3 Step one: searching for survey names

The search terms are the keyword list the LSMS team already used in its Google Alerts tracker, 109 terms across 10 survey families. A few entries were merged or dropped as duplicates, for example “The Uganda National Panel survey”, which could never match anything that the shorter “Uganda National Panel Survey” did not already catch.

These names differ enormously in how ambiguous they are. “Uganda National Panel Survey” cannot plausibly appear outside LSMS-related work. The acronym “IHS” appears in thousands of unrelated papers about health plans and hospital systems. And some are real survey titles whose words describe the survey rather than identify it: “National Panel Survey” belongs to both Tanzania and Uganda, and almost every agency ran a “High-Frequency Phone Survey” during COVID. One rule for all three would either discard real papers or admit irrelevant ones, so **each keyword carries one of three tiers**:

Tier	Terms	Example	Matching rule
A	38	“Uganda National Panel Survey”, “LSMS-ISA”, “EHCVM 2018”	Unambiguous: the name contains a country, or an acronym that cannot plausibly mean anything else. A full-text match is accepted on its own. No country context and no field filter are required. Most genuine papers name their data source in the Methods section rather than the abstract, and OpenAlex indexes full text, so this is the tier that does most of the work.
B	28	“National Panel Survey”, “High-Frequency Phone Survey”, “Living Standards Survey”	A real survey name, but a generic one: the words describe the survey rather than identify it, and other countries or programmes use the same phrase. Requires an LSMS country word somewhere in the document AND an allowed subject field. Case-insensitive: casing carries no information in a phrase.
C	43	“IHPS”, “LSMS”, “UNPS”, “ESS4”	Short acronym. Same country and field gating as B, plus a case check: if the acronym appears in the title or abstract with the wrong casing (“LSMs”, the plural of land-surface model) that is a positive sign of a collision and the paper is dropped. If it does not appear there at all we cannot tell, and the query’s own country gate stands.

2.4 Keeping the search precise

Search engines normally treat related words as equivalent, so a search for LSMS also returns LSM, the standard abbreviation for land-surface model. That single behaviour returned **13,506 results for one fiscal year where the correct number was 698**.

Searches therefore run in exact mode, which switches that off. Exact mode still ignores capitalisation, so “LSMs” and “LSMS” remain identical to the search engine, which is what the Tier C case check in the table above is for.

2.5 Step two: separating genuine use from a passing mention

Finding a survey name proves only that the words appear somewhere in the paper. Step two therefore asks two different questions about every collected paper and scores them separately:

Is this really one of our surveys? A name can collide with an unrelated acronym, or belong to another country’s survey that happens to share a phrase. This score is called **identity**.

Did the authors actually work with the data? Or do they only mention it in passing. This score is called **use**.

Each piece of evidence is worth 2 points if it is strong and 1 point if it is weak. A paper is counted only if it scores at least 2 for identity **and** at least 2 for use. So one strong piece of evidence settles a question on its own, two weak ones together also settle it, and a single weak one never does.

Scoring the two questions separately is the point of the design. Added into one number, a paper could pass on name recognition alone: mention the survey three times, add a World Bank co-author, and a literature review would outrank a real empirical study. Keeping them apart makes **“mentions the survey but never uses the data” unable to pass at all**, which the previous version could not prevent.

The evidence used for each question is listed below.

Axis	Signal	Weight
Identity	A known survey name appears in the title	Strong (2)
Identity	A known survey name appears in the abstract	Strong (2)
Identity	Admitted on a Tier A (unambiguous) name	Strong (2)
Identity	Admitted only on a Tier B/C (generic or acronym) term	Weak (1)
Identity	Matched terms from two or more survey families	Strong (2)
Identity	Matched two or more term variants of one survey	Weak (1)
Identity	An LSMS country is named in the title or abstract	Weak (1)
Identity	The LSMS program is named in the abstract	Weak (1)
Identity	A World Bank-affiliated author	Weak (1)
Use	Abstract names the survey AND uses explicit first-person data-use language (“we use”, “using data from”, “microdata”)	Strong (2)
Use	Abstract names the survey AND uses weaker data-source language	Weak (1)
Use	Full text: first-person data-use language within 40 words of the survey name (also grants +1 identity)	Strong (2)
Use	Full text: data-object language near the survey name	Weak (1)
Use	Full text: who ran, funded or implemented the survey, near its name	Weak (1)
Use	Untied evidence: empirical vocabulary, or data-use language with no survey named	Capped at 1 total

As for the Use score, evidence has to point at **our** survey specifically. An abstract saying “we use household survey data” shows the authors used some data, not that they used this one. Evidence counts in full only when it is attached to the survey: either the two appear within 40 words of each other in the full text, or the abstract names the survey and describes using data in the same place. Unattached evidence, such as general statistical vocabulary, is capped at 1 point so it can support a real finding but never stand in for one.

Because abstracts are short and often say nothing about data sources, a second pass searches the full text of any paper whose use score is still short. It asks whether the survey name appears within 40 words of language describing working with the data. Note that this pass is triggered by a low **use** score specifically, not a low total: a paper can carry overwhelming evidence that it is the right survey and still show nothing about whether the data was used, and that is exactly the paper worth checking.

The phrases it searches for were chosen by measurement rather than intuition. Every candidate phrase was tested against a 36-paper set that had already been assessed by hand, and scored on how many genuine users against how many non-users it caught. “Using data from” separated cleanly, catching 8 real users and 1 non-user. “This paper uses” and “based on the” proved to be **negative** indicators, appearing more often in citations than in real use, and were removed. Citing the World Bank microdata catalogue scored 2 against 2, no separating power at all, so it is recorded on the paper as a note but earns no points.

2.6 Rules that override the score

Two rules sit outside the scoring. Six publication types can never be an empirical use of survey data and are excluded regardless of score: conference abstract, dataset, paratext, erratum, letter and software. Editorials are deliberately not on that list.

Literature reviews are held to a higher standard rather than excluded. A review that genuinely re-analyses the data still counts, but it must show one strong piece of attached evidence rather than an accumulation of weak ones, because reviews are the largest single category of papers that name a survey without using it.

Every piece of evidence that applied is written to the workbook, along with both scores, so any decision can be traced back to its reasons.

2.7 Removing duplicate records

Before anything is counted, candidate papers are checked against each other by normalized title and, where available, DOI. Exact matches are merged automatically.

2.8 Fiscal year and author geography

- Rows on the Papers sheet are sorted by World Bank Group fiscal year (July 1 to June 30) in descending order (two-digit FY are sorted accounting for the fact that years 80-99 come before 01-27). Then, the papers are sorted alphabetically within each FY.
- Author geography is inferred from each author's institutional affiliation data returned by OpenAlex, mapped to country codes and checked against the eight ISA countries and the wider Sub-Saharan Africa list.

2.9 What the workbook contains

A single Excel workbook with the following sheets, in this order:

Sheet	Contents
Papers	One row per discovered paper: title, DOI, authors, institutional affiliations, publication year/FY, dataset country, research topic, peer-reviewed / working-paper flag, journal tier, the identity and use scores, and the individual signals that fired (relevance flags).
Analysis	The headline FLOW and Africa-SHARE numbers, a breakdown by match tier (strongest tier per paper, so the three sum to 100%) and by the identity/use axes, plus the summary charts. Charts sit near the top of the sheet.
FY Trend	A bar chart and a line-with-markers chart of papers per WBG fiscal year, both with data labels. The current, still-open FY is excluded so the trend isn't distorted by a partial year.
Keywords	All 109 keywords, each with its tier, the matching rule applied, the required country context words, and the exact OpenAlex query sent, so any result can be traced back to the search that found it.
Search Log	An audit trail: every query that was run, when, and how many raw results it returned, before filtering.
Not Relevant (Backup)	Papers that matched a keyword but scored 0 or 1 on relevance. Kept for reference, but excluded from every headline count and chart. Always the last sheet.

2.10 Known limits

Two kinds of error remain, and both follow from the same constraint. OpenAlex allows the pipeline to ask whether two phrases appear near each other in a paper, but it does not hand over the text itself. The pipeline therefore cannot tell which part of a paper a phrase came from, and a name in the reference list looks the same to it as a name in the methods section.

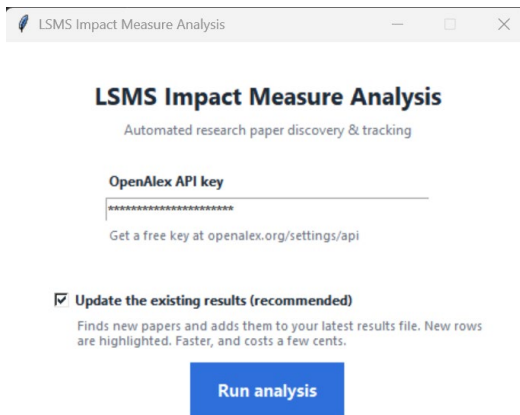
Reference-only matches. A paper that cites a survey without using it still contains the survey name, and a dataset citation carries collection language of its own, for example “Uganda National Panel Survey 2019-2020, collected by Uganda Bureau of Statistics”. That is indistinguishable from a methods sentence when only the distance between words is visible. Four papers in the review set are counted this way when they should not be.

Unindexed data statements. Some papers declare their data in a table or a dataset section that OpenAlex has not indexed. In one verified case the phrase “Living Standards Measurement Study” appears in the paper's own dataset section yet is absent from the index at every query length and proximity setting, while the acronym alone is present. No query formulation reaches that paper. Section-aware analysis would address both, and was tested. Europe PMC returns sectioned full text and separates use from mention correctly where it has coverage, but it indexes none of the economics venues this corpus depends on, covering 4 of 10 sample papers and misclassifying one genuine user. **Closing these gaps requires a full-text source with reliable coverage of development and agricultural economics, not further tuning of the current signals.**

3. Program Interface (when running “LSMS Impact Analysis.bat”)

Starting screen

Where the OpenAlex API key is entered, the update-or-full choice is set, and the analysis is started.



LSMS Impact Measure Analysis

Automated research paper discovery & tracking

OpenAlex API key

Get a free key at openalex.org/settings/api

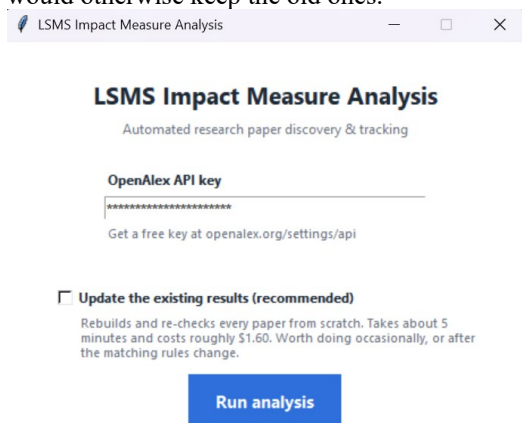
☒ **Update the existing results (recommended)**

Finds new papers and adds them to your latest results file. New rows are highlighted. Faster, and costs a few cents.

Run analysis

"Update the existing results (recommended)" checkbox

Ticked, the default shown above: an update run. It still searches everything, but only pays to re-check papers that were not already scored, so it finds new papers and adds them to the latest results file with the new rows highlighted. Faster, and costs a few cents. Unticked: a full run. It rebuilds and re-checks every paper from scratch, which takes about 5 minutes and costs roughly \$1.60. Worth doing occasionally, or after the matching rules in keywords.py change, since an update run reuses stored verdicts and would otherwise keep the old ones.



LSMS Impact Measure Analysis

Automated research paper discovery & tracking

OpenAlex API key

Get a free key at openalex.org/settings/api

☐ **Update the existing results (recommended)**

Rebuilds and re-checks every paper from scratch. Takes about 5 minutes and costs roughly \$1.60. Worth doing occasionally, or after the matching rules change.

Run analysis

While it's running

A blue spinner runs while the program searches OpenAlex and builds the workbook (about 5 minutes).

LSMS Impact Measure Analysis

Automated research paper discovery & tracking



Running full analysis...

Searching OpenAlex and rebuilding the workbook from scratch. Usually about 5 minutes.

Do not close this window while the analysis is running.

When it's finished

A green checkmark appears with the name of the finished workbook and an Open File button.

LSMS Impact Measure Analysis

Automated research paper discovery & tracking



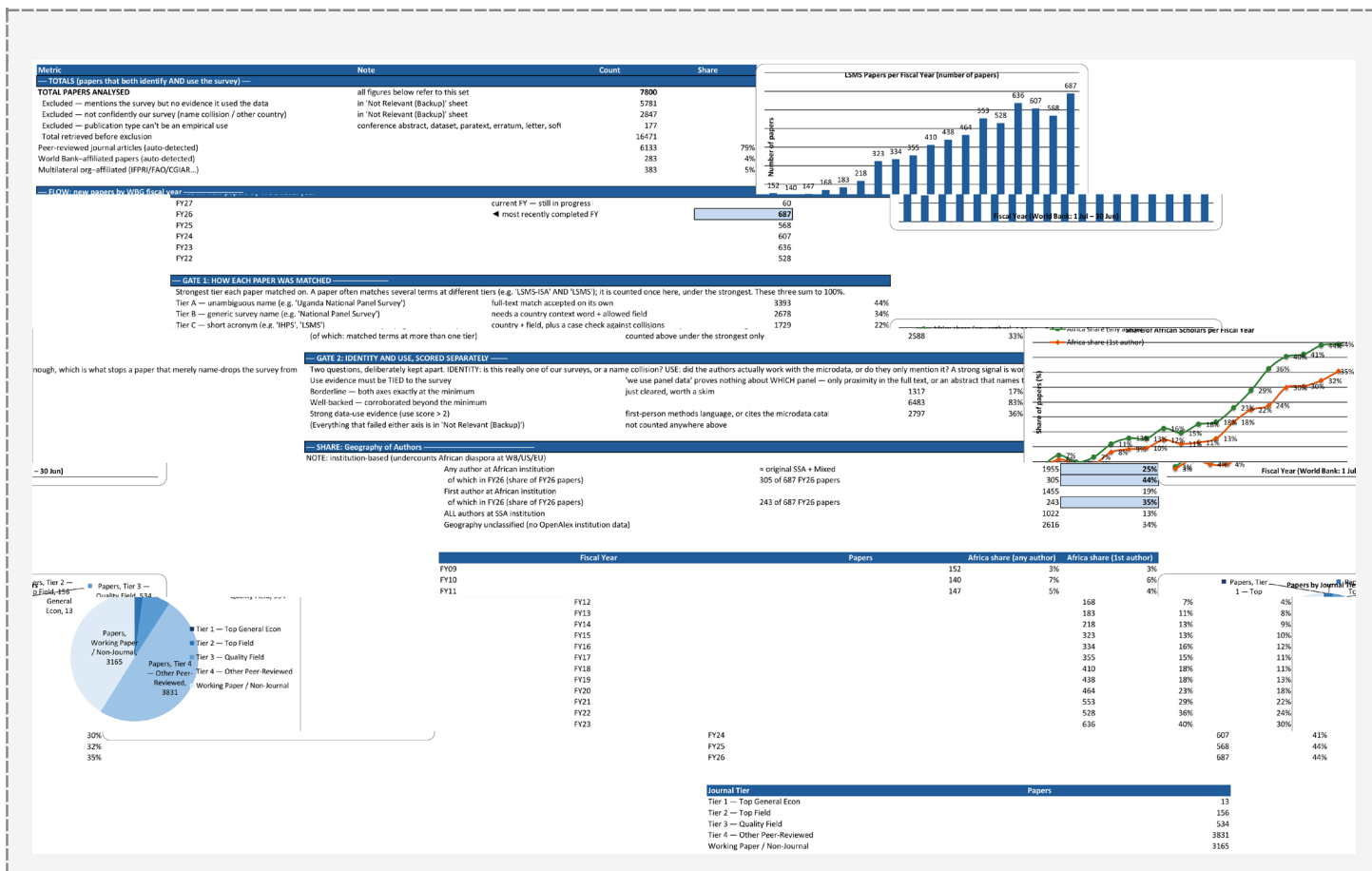
Ready

LSMS_papers_20260713_115457.xlsx

Open File

[Run again](#)

Analysis



Papers

title	doi	year	month	pub_date	ly	publication_type	pr	pub_type	journal_title	pr_review	venue	authors	first_author	second_author	n_authors	link	open	
Agricultural extension increases farm productivity but not women	https://doi.org/10.1080/23311932.2020.1820210	2026	7	2026-07-05	PV27	article	Journal Article 4 – Other Peer Yes		Cogent Food & Agriculture	Monica Sebastian Kaur	Monica Sebastian Kaur	1	https://www.tandfonline.com/doi/full/10.1080/23311932.2020.1820210			1	https://www.tandfonline.com/doi/full/10.1080/23311932.2020.1820210	TF
CRC - Vietnam	https://doi.org/10.5281/zenodo.1191200	2026	7	2026-07-01	PV27	dataset	Working Paper WP – Working No		Zenodo (ZEN Europe)	On Thanh Cong Nguyen	Thanh Cong Nguyen	1	https://zenodo.org/record/1191200/files/CRC-Vietnam.pdf			1	https://zenodo.org/record/1191200/files/CRC-Vietnam.pdf	TF
Drought shocks and the adoption of multiple sustainable intensification practices in smallholder farms in Malawi	https://doi.org/10.1007/s10662-020-02061-0	2026	7	2026-07-02	PV27	article	Journal Article 4 – Other Peer Yes		Agricultural Economics	Abhay Gehringhwal, Girellabhai G. Gehringhwal, Girellabhai G. Gehringhwal	Abhay Gehringhwal, Girellabhai G. Gehringhwal, Girellabhai G. Gehringhwal	4	https://doi.org/10.1007/s10662-020-02061-0			4	https://doi.org/10.1007/s10662-020-02061-0	TF
Economic viability of loan-financed urban residential solar PV systems in India	https://doi.org/10.1177/014458972036124	2026	7	2026-07-06	PV27	article	Journal Article 4 – Other Peer Yes		Energy Exploration & Exploitation	Stephen Afana-Mensah, Flavio Odilo Y. Mensah	Stephen Afana-Mensah, Flavio Odilo Y. Mensah	5	https://doi.org/10.1177/014458972036124			5	https://doi.org/10.1177/014458972036124	TF
Effect of Household Size on Educational Outcomes in Tanzania: A Case Study of the Morogoro Region	https://doi.org/10.52589/ajes-0206-01	2026	7	2026-07-06	PV27	article	Journal Article 4 – Other Peer Yes		African Journal of Economics & Statistics	S. Seleman, L. M. Odiano	S. Seleman, L. M. Odiano	2	https://doi.org/10.52589/ajes-0206-01			2	https://doi.org/10.52589/ajes-0206-01	TF
Estimating the Returns to Mobile Technology in Smallholder Agri	https://doi.org/10.22094/ag-econ-404	2026	7	2026-07-02	PV27	other	Repository / VWP – Working No		AgEcon Search (University of Nebraska-Lincoln)	Nomore Nkhoo Kianon Chen	Nomore Nkhoo Kianon Chen	2	https://doi.org/10.22094/ag-econ-404			2	https://doi.org/10.22094/ag-econ-404	TF
PervasiveSafes2025	https://doi.org/10.5281/zenodo.1191200	2026	7	2026-07-01	PV27	dataset	Repository / VWP – Working No		Swedish National Data Science Center	Seamante Nihoma, Jeanette Johansson	Seamante Nihoma, Jeanette Johansson	2	https://doi.org/10.5281/zenodo.1191200			2	https://doi.org/10.5281/zenodo.1191200	TF
Predicting curbedile food waste generation from responses to a survey	https://doi.org/10.1016/j.wasman.2020.10.016	2026	7	2026-07-01	PV27	article	Journal Article 4 – Other Peer Yes		Waste Management	Brână R. Rău, Li. Xu, Gaby K. Martin, J. G. Rău	Brână R. Rău, Li. Xu, Gaby K. Martin, J. G. Rău	4	https://doi.org/10.1016/j.wasman.2020.10.016			4	https://doi.org/10.1016/j.wasman.2020.10.016	TF
Solar Use Dynamics in Uganda: Household Characteristics and En	https://doi.org/10.32479/jeep.24152	2026	7	2026-07-05	PV27	article	Journal Article 4 – Other Peer Yes		International Journal of En	Grace Alinafne, Florence Banu	Grace Alinafne, Florence Banu	2	https://doi.org/10.32479/jeep.24152			2	https://doi.org/10.32479/jeep.24152	TF
Weather Risk and Crop Switching in Perennial Agriculture: Eviden	https://doi.org/10.22004/ag-econ-404	2026	7	2026-07-12	PV27	other	Repository / VWP – Working No		AgEcon Search (University of Nebraska-Lincoln)	Hannah Kim	Hannah Kim	1	https://doi.org/10.22004/ag-econ-404			1	https://doi.org/10.22004/ag-econ-404	TF
When sellers become buyers: Food inflation, seasonal market p	https://doi.org/10.22004/ag-econ-404	2026	7	2026-07-12	PV27	other	Repository / VWP – Working No		AgEcon Search (University of Nebraska-Lincoln)	Divine-Love Akam	Divine-Love Akam	1	https://doi.org/10.22004/ag-econ-404			1	https://doi.org/10.22004/ag-econ-404	TF
Who pays for fertilizer–input market power? A CGE-microsimulat	https://doi.org/10.17625/ajae.2020.10.016	2026	7	2026-07-08	PV27	dataset	Repository / VWP – Working No		Manuscript Data	Patrice Zidomaba	Patrice Zidomaba	1	https://doi.org/10.17625/ajae.2020.10.016			1	https://doi.org/10.17625/ajae.2020.10.016	TF
Who Pays for Fertilizer? Input Market Power in Smallholder Agri	https://doi.org/10.17625/ajae.2020.10.016	2026	7	2026-07-08	PV26	dataset	Journal Article 4 – Other Peer Yes		Manuscript Data	Patrice Zidomaba, Sijie Hu	Patrice Zidomaba, Sijie Hu	2	https://doi.org/10.17625/ajae.2020.10.016			2	https://doi.org/10.17625/ajae.2020.10.016	TF
Who Pays for Fertilizer? Input Market Power in Smallholder Agri	https://doi.org/10.17625/ajae.2020.10.016	2026	7	2026-07-08	PV26	dataset	Journal Article 4 – Other Peer Yes		Manuscript Data	Patrice Zidomaba, Sijie Hu	Patrice Zidomaba, Sijie Hu	2	https://doi.org/10.17625/ajae.2020.10.016			2	https://doi.org/10.17625/ajae.2020.10.016	TF
Who Pays for Fertilizer? Input Market Power in Smallholder Agri	https://doi.org/10.17625/ajae.2020.10.016	2026	7	2026-07-08	PV26	dataset	Journal Article 4 – Other Peer Yes		Manuscript Data	Patrice Zidomaba, Sijie Hu	Patrice Zidomaba, Sijie Hu	2	https://doi.org/10.17625/ajae.2020.10.016			2	https://doi.org/10.17625/ajae.2020.10.016	TF
Who Pays for Fertilizer? Input Market Power in Smallholder Agri	https://doi.org/10.17625/ajae.2020.10.016	2026	7	2026-07-08	PV26	dataset	Journal Article 4 – Other Peer Yes		Manuscript Data	Patrice Zidomaba, Sijie Hu	Patrice Zidomaba, Sijie Hu	2	https://doi.org/10.17625/ajae.2020.10.016			2	https://doi.org/10.17625/ajae.2020.10.016	TF
Who Pays for Fertilizer? Input Market Power in Smallholder Agri	https://doi.org/10.17625/ajae.2020.10.016	2026	7	2026-07-08	PV26	dataset	Journal Article 4 – Other Peer Yes		Manuscript Data	Patrice Zidomaba, Sijie Hu	Patrice Zidomaba, Sijie Hu	2	https://doi.org/10.17625/ajae.2020.10.016			2	https://doi.org/10.17625/ajae.2020.10.016	TF
Who Pays for Fertilizer? Input Market Power in Smallholder Agri	https://doi.org/10.17625/ajae.2020.10.016	2026	7	2026-07-08	PV26	dataset	Journal Article 4 – Other Peer Yes		Manuscript Data	Patrice Zidomaba, Sijie Hu	Patrice Zidomaba, Sijie Hu	2	https://doi.org/10.17625/ajae.2020.10.016			2	https://doi.org/10.17625/ajae.2020.10.016	TF
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Who Pays for Fertilizer? Input Market Power in Smallholder Agri	https://doi.org/10.17625/ajae.2020.10.016	2026	7	2026-07-08	PV26	dataset	Journal Article 4 – Other Peer Yes		Manuscript Data	Patrice Zidomaba, Sijie Hu	Patrice Zidomaba, Sijie Hu	2	https://doi.org/10.17625/ajae.2020.10.016			2	https://doi.org/10.17625/ajae.2020.10.016	TF
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Who Pays for Fertilizer? Input Market Power in Smallholder Agri	https://doi.org/10.17625/ajae.2020.10.016	2026	7	2026-07-08	PV26	dataset	Journal Article 4 – Other Peer Yes		Manuscript Data	Patrice Zidomaba, Sijie Hu	Patrice Zidomaba, Sijie Hu	2	https://doi.org/10.17625/ajae.2020.10.016			2	https://doi.org/10.17625/ajae.2020.10.016	TF
Who Pays for Fertilizer? Input Market Power in Smallholder Agri	https://doi.org/10.17625/ajae.2020.10.016	2026	7	2026-07-08	PV26	dataset	Journal Article 4 – Other Peer Yes		Manuscript Data	Patrice Zidomaba, Sijie Hu	Patrice Zidomaba, Sijie Hu	2	https://doi.org/10.17625/ajae.2020.10.016			2	https://doi.org/10.17625/ajae.2020.10.016	TF
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Who Pays for Fertilizer? Input Market Power in Smallholder Agri	https://doi.org/10.17625/ajae.2020.10.016	2026	7	2026-07-														

5. User Guide -- Setup Instructions

This is the full path from receiving the zip file to opening a finished workbook. It only needs to be done in full once per computer; after first use, just follow Step 4.

Step 1: Unzip the folder

Extract the zip file anywhere convenient (Desktop or Documents both work). Keep all the files together in the same folder.

Step 2: Run “Initial Setup.bat” (once)

If you don’t have Python installed, run Initial Setup.bat. This is a one-time step that prepares the computer to run the pipeline:

1. It checks whether Python is already installed.
2. If not, it downloads Python 3.12 directly from python.org and installs it in the background.
3. It then installs the handful of Python packages the pipeline needs (from requirements.txt).
4. When it says “Setup Complete,” you can close the window.

This step needs an internet connection and usually takes one to two minutes (longer if Python itself has to be downloaded and installed).

Step 3: Get a free OpenAlex API key

1. Go to openalex.org and create a free account (any email works, including a WBG email).
2. Once logged in, go to openalex.org/settings/api and copy your personal API key.
3. Keep it somewhere safe, you need to paste it into the program once, and it will remember it for every run after that.

Note: No credit card or payment method is required for normal use. A free OpenAlex API key comes with roughly \$5.00 of free daily usage. A full run costs about \$1.60; an update run (the default, and what most people should use) costs only a few cents, since it re-checks just the papers that were not already scored. That covers several runs a day for most use. If a heavier day of full runs needs more than the daily allowance, register a second API key or buy usage credits on the OpenAlex API website.

Step 4: Run “LSMS Impact Analysis.bat”

1. Double-click LSMS Impact Analysis.bat
2. The first time, paste your OpenAlex API key into the field and click Run analysis.
3. A blue spinner shows while the analysis runs--usually about 5 minutes. Leave the window open.
4. When it finishes, a green checkmark appears with the name of the new Excel file and an Open File button. click it to open the workbook directly.

6. Public Dashboard

Every export also mirrors itself onto a public, no-login dashboard, so the results can be checked from a browser without opening Excel:

https://bdornyei42.github.io/WBG_LSMS_Impact/

The page is built by `build_site.py`, which reads the newest `LSMS_papers_*.xlsx` in the project folder and writes `docs/data.json` and `docs/papers.json` from it. It reproduces the Analysis sheet as an interactive page, plus a searchable version of the Papers sheet:

- The three summary charts: papers per fiscal year, share of African-affiliated authorship, and papers by journal tier.
- The same Totals, match-tier (Gate 1), identity/use (Gate 2), and geography breakdowns as the workbook's Analysis sheet.
- A searchable, sortable table of every paper in the Papers sheet, with links to each paper.

It updates when "Save changes to GitHub" (`save.bat`) is run: that script calls `build_site.py` to regenerate `docs/data.json` and `docs/papers.json` from the latest export, then commits and pushes `docs/` along with the rest of the repository. GitHub Pages serves whatever is in `docs/` on the main branch directly, so the page goes live within a minute or two of the push, with no separate deploy step. Running `discover.py` or the LSMS Impact Analysis.bat launcher on its own does not update the dashboard by itself; `save.bat` is the step that publishes a run's results.